AI-Powered Sanskrit NLP Framework for Morphological and Semantic Analysis

IT Project Management (21MGH303P)

B.Tech – III Year | 6th Semester

(AI-driven framework for morphological analysis, semantic parsing, and structured knowledge extraction from classical Sanskrit texts based on Paninian grammar.)

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Problem Statement

- **Current Challenges in Sanskrit NLP**
  - Lack of integrated systems combining Paninian grammar with modern AI techniques
  - Existing tools are primarily:
    - Rule-based (limited scalability)
    - Or statistical (lack linguistic depth)
- **Technical Limitations**
  - Inefficient morphological analysis for complex Sanskrit structures
  - Absence of robust semantic parsing frameworks
  - Limited capability for structured knowledge extraction
- **Research & Practical Gaps**
  - Minimal support for large-scale corpus processing
  - Poor accessibility for researchers (lack of standardized APIs)
  - High dependency on manual linguistic analysis

Project Objectives

Primary Objectives

- Develop a scalable NLP framework for Sanskrit text analysis
- Ensure integration of Paninian grammar with AI techniques

Management Objectives

- Deliver project within:
- Budget: ₹33–41 Lakhs
- Timeline: 218 Days
- Maintain quality standards (≥92% accuracy target)

Operational Objectives

- Provide API-based access for researchers
- Enable large-scale corpus processing
- Ensure ease of use and adoption

Strategic Goal

- Establish a sustainable and extensible platform for academic and research use

Requirements & Scope Overview

Requirement Approach

- **Requirements gathered via:**
  - Stakeholder workshops
  - Document analysis
  - Institutional survey

Key Requirement Categories

- Functional: Core system capabilities (analysis, API access, processing)
- Non-Functional: Performance, accuracy, scalability, reliability

Scope Definition

In Scope

- NLP framework development
- API deployment and documentation
- Cloud-based processing

Out of Scope

- Mobile application
- Real-time streaming systems
- OCR / speech processing

Priority Management

- Requirements prioritized using MoSCoW method
- Focus on critical deliverables for initial release

Stakeholder Management

Key Stakeholders

- Sponsors (UGC / MHRD)
- Sanskrit Linguists
- NLP Engineers & Data Team
- Researchers & End Users
- Cloud Service Provider

Communication Strategy

- Weekly reviews (core team)
- Monthly reporting (sponsors)
- Feedback cycles (end users)

Management Approach

- Stakeholders classified based on Power vs Interest
- Engagement strategies:
  - High Power–High Interest → Close monitoring
  - Others → periodic communication

Outcome

- Improved decision-making
- Reduced project risks
- Better stakeholder alignment

Stakeholder Classification Framework

Stakeholders are mapped based on:

- Power (influence on project)
- Interest (level of involvement)

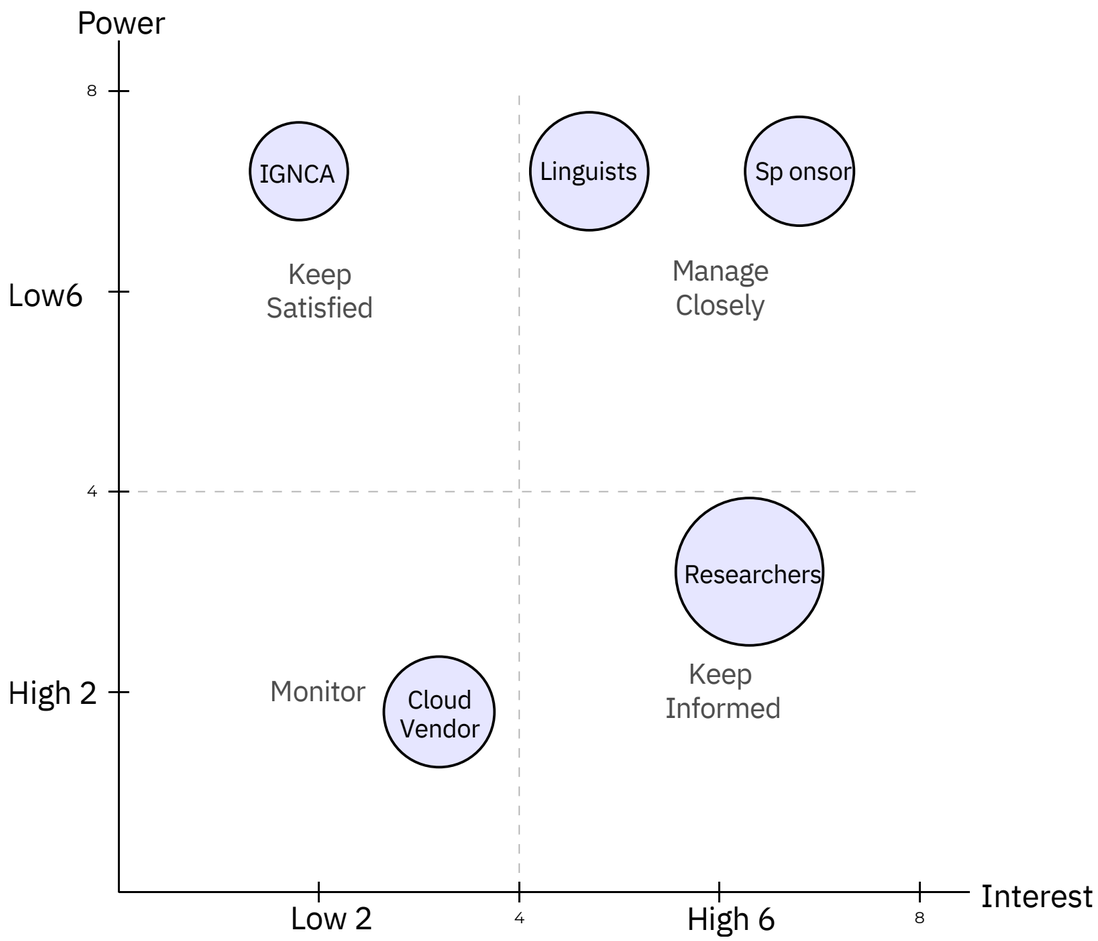
Management Strategy

- **High Power – High Interest:** Manage closely
- **High Power – Low Interest:** Keep satisfied
- **Low Power – High Interest:** Keep informed
- **Low Power – Low Interest:** Monitor

Project Insight

- **Critical stakeholders (Sponsors, Linguists) require continuous engagement**
- **End users require regular updates and feedback integration**

Stakeholder Power/Interest Grid



Requirement Prioritization (MoSCoW)

Purpose

- Ensure focused **delivery within time** and budget constraints
- Prioritize requirements based on **business value and criticality**

Categories

- **Must Have:** Essential for project success
- **Should Have:** Important but not critical
- **Could Have:** Value-adding enhancements
- **Won't Have:** Out of current scope

Management Insight

- Enables **controlled scope and prevents scope creep**
- Supports **phased delivery** strategy

Requirement Prioritization (MoSCoW)

Must Have - Morphological Analyzer - Semantic Parser - Core NLP Engine	Should Have - REST API - Batch Processing - Knowledge Extraction
Could Have - UI Visualization - Bilingual Interface	Won’t Have -OCR - Speech Processing - Translation